GradientCIO

INVESTMENT DUE DILIGENCE

Northstar Select Equity Composite

Single-manager performance assessment · Investment Committee review

REVIEW
Single manager

OBSERVATION WINDOW
2020-01-31 to 2025-12-31

OBSERVATIONS
72 monthly

PRIMARY BENCHMARK
Northstar Core Equity Benchmark

01 Executive conclusion

Positive full-window edge with moderate risk-adjusted reliability

ANNUALIZED RETURN	EXCESS VS BENCHMARK	INFORMATION RATIO	MAX DRAWDOWN
6.5%	+1.4pp	0.32	-18.2%
2020-01-31 to 2025-12-31 S1	Annualized, same 72 months S2	2020-01-31 to 2025-12-31 S2	Observed through 2025-12-31 S1

Positive observed edge; reliability remains moderate.

The manager outpaced the matched benchmark across every returned start-date window, but the 0.32 information ratio and 72-month sample do not establish persistence. A negative rolling 36-month excess path or downside capture above 100% would weaken the view S2 S3.

Key judgment — full-window outcome

Observation: The manager returned 6.5% annualized versus 5.1% for the matched benchmark from 2020-01-31 through 2025-12-31 S1. Why it matters: The observed track record compounded to a higher ending value. Uncertainty: Only 72 monthly observations cover one severe market decline. What would change the view: A governed 84-month common-window refresh with annualized excess at or below 0.0 percentage points.

Key judgment — relative efficiency

Observation: Annualized excess return was 1.4 percentage points with a 0.32 information ratio S2. Why it matters: Positive relative return was achieved with only moderate efficiency. Uncertainty: The 72-observation tracking-error estimate may shift as history extends. What would change the view: A governed information ratio below zero or above 0.50 over at least 84 months.

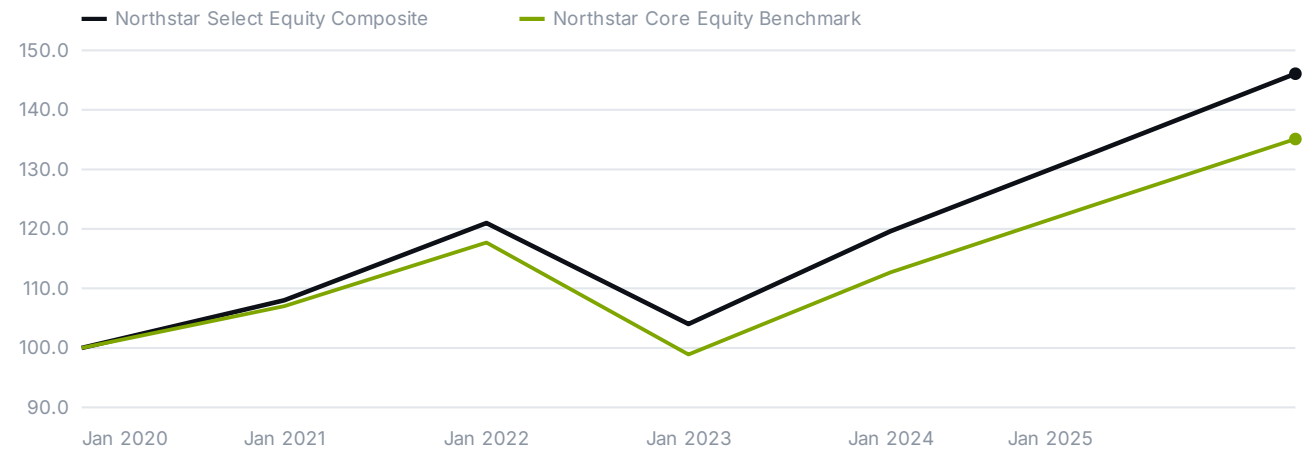
Key judgment — benchmark dependence

Observation: Excess return narrows to 0.5 percentage points against the advisory alternative benchmark S5. Why it matters: The sign remains positive, but the strength of the relative conclusion depends on benchmark choice. Uncertainty: The alternative benchmark is not the canonical server match. What would change the view: Mandate confirmation that replaces the primary match or returns negative alternative-benchmark excess.

02 Track record over time

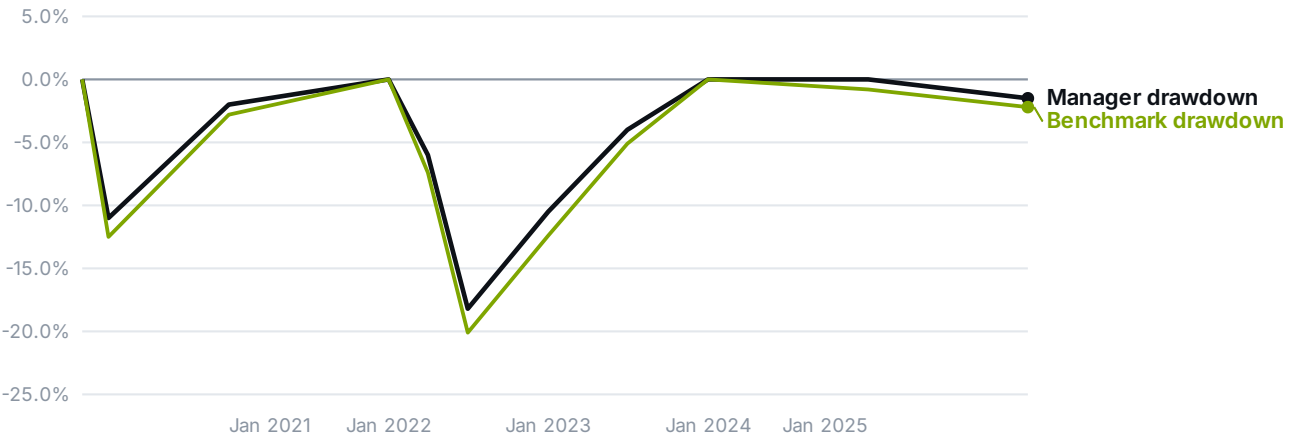
The ending wealth advantage survived the deepest observed drawdown

CUMULATIVE GROWTH OF 100 · 2019-12-31 TO 2025-12-31 [S2]



Returned cumulative-growth observations; no local interpolation S2.

OBSERVED DRAWDOWN PATH · 2020-01-31 TO 2025-12-31 [S2]



Returned peak-to-trough observations S2.

Analysis — Ending wealth advantage survived drawdown

Observation: Growth ended at 146.1 versus 135.1, with drawdowns of -18.2% and -20.1% S2. Why it matters: The wealth advantage survived the deepest loss. Uncertainty: Returned annual display points omit monthly path detail. What would change the view: An ending value below 135.1 or manager drawdown below -20.1%.

03 Data and benchmark coverage

Complete monthly coverage with a server-matched primary benchmark

EVIDENCE COVERAGE

Manager return series	**Complete**	72 of 72 monthly observations, 2020-01-31 to 2025-12-31 S1
Primary benchmark	**Complete**	Northstar Core Equity Benchmark, server matched S1
Relative and capture views	**Complete**	Common 72-month window S2
Date-window robustness	**Complete**	Four returned start dates S3
Alternative benchmark	**Partly available**	Governed comparison returned; mapping remains advisory S5

SERIES AND BENCHMARK IDENTITY

Manager series	**returns-test-northstar-select** S1
Primary benchmark series	**benchmark-test-northstar-core-equity** S1
Frequency / observations	**Monthly / 72** S1
Return basis	**Total return, net of reported manager fees** S1
Currency	**USD** S1
Alternative benchmark	**Northstar Broad Equity Benchmark, advisory** S5

Coverage boundary

The report uses the common returned window and preserves the server-matched primary benchmark. The advisory alternative benchmark is isolated in its own sensitivity section and does not replace the primary result S1 S5.

04 Absolute risk and return

Positive compounding came with double-digit volatility and an 18.2% drawdown

CALENDAR-YEAR RETURNS · GROUPED MANAGER AND BENCHMARK BARS [S2]



Adjacent bars form each calendar-year manager/benchmark group S2.

CALENDAR-YEAR RETURN DETAIL [S2]

YEAR	MANAGER	BENCHMARK	EXCESS	SOURCE
2020	8.0%	7.0%	+1.0pp	S2
2021	12.0%	10.0%	+2.0pp	S2
2022	-14.0%	-16.0%	+2.0pp	S2
2023	15.0%	14.0%	+1.0pp	S2
2024	11.0%	10.0%	+1.0pp	S2
2025	10.0%	9.0%	+1.0pp	S2

REALIZED MONTHLY RETURN DISTRIBUTION [S2]



Returned distribution of the 72 realized monthly manager returns S2.

MODEL OUTPUTS (SIMULATED)

10-YEAR OUTPUT	RETURNED VALUE	SOURCE
Expected annualized return	6.8%	S4
Expected annualized volatility	12.1%	S4
10-year terminal wealth median	193.4	S4

Server-returned 10-year simulated outputs from the outer response; not realized series-view evidence S4.

RISK AND RETURN BY COMMON WINDOW [S1]

WINDOW	SERIES	ANN. RETURN	ANN. VOLATILITY	SHARPE	MAX DRAWDOWN	SOURCE
2020-01-31 to 2025-12-31	Manager	6.5%	12.4%	0.45	-18.2%	S1
2020-01-31 to 2025-12-31	Benchmark	5.1%	11.8%	0.36	-20.1%	S1
2021-01-31 to 2025-12-31	Manager	6.2%	12.8%	0.41	-18.2%	S1
2021-01-31 to 2025-12-31	Benchmark	4.8%	12.1%	0.32	-20.1%	S1
2023-01-31 to 2025-12-31	Manager	12.0%	10.9%	0.92	-7.1%	S1
2023-01-31 to 2025-12-31	Benchmark	11.0%	10.5%	0.86	-8.0%	S1

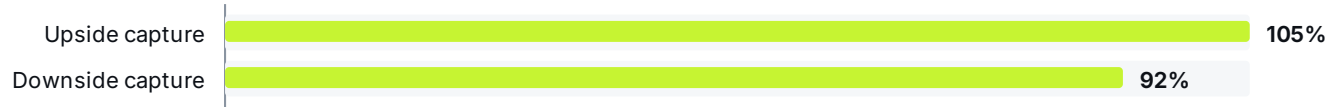
Analysis — Return retained absolute risk

Observation: The manager returned 6.5% with 12.4% volatility and -18.2% drawdown S1. Why it matters: Positive compounding required material loss tolerance. Uncertainty: Seventy-two months include one major drawdown. What would change the view: An 84-month governed update below 5.1% return or below -20.1% drawdown.

05 Relative performance

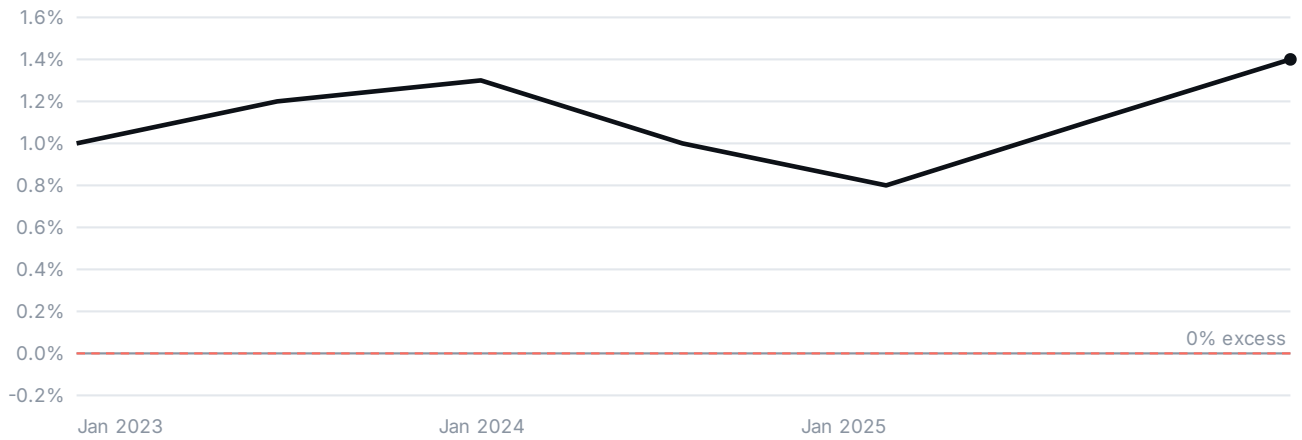
Capture was favorable, while the information ratio remained moderate

UPSIDE AND DOWNSIDE CAPTURE · 2020-01-31 TO 2025-12-31 [S2]



Benchmark up/down month classifications returned by the relative-performance view S2.

ROLLING 36-MONTH ANNUALIZED EXCESS RETURN · RELATIVE VIEW [S2]



Repeated here to pair path evidence with the realized relative table S2.

REALIZED RELATIVE PERFORMANCE [S2]

WINDOW	MANAGER	BENCHMARK	EXCESS	TRACKING ERROR	INFORMATION RATIO	SOURCE
6 years	6.5%	5.1%	+1.4pp	4.3%	0.32	S2
5 years	6.2%	4.8%	+1.4pp	4.5%	0.31	S2
3 years	12.0%	11.0%	+1.0pp	3.8%	0.26	S2

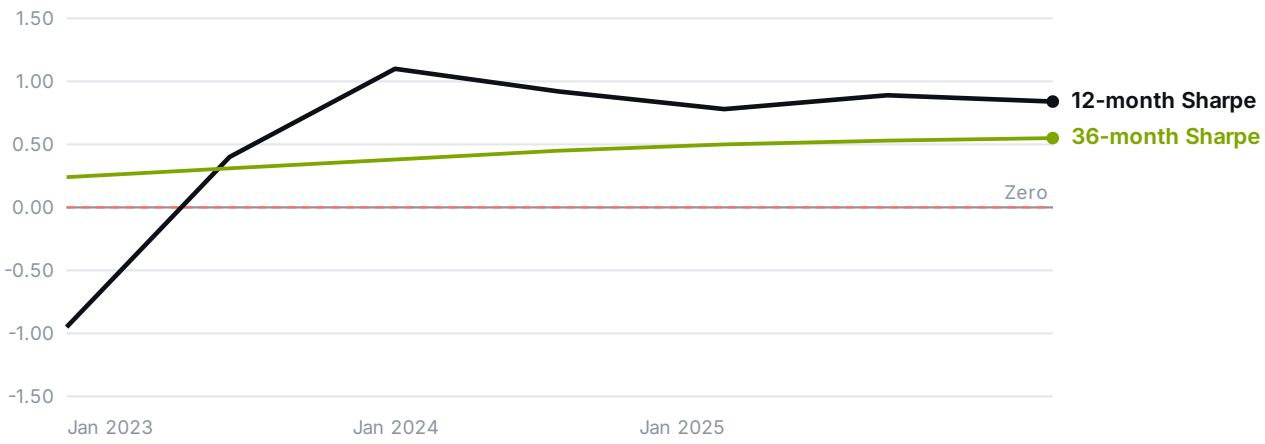
Analysis — Positive edge, moderate efficiency

Observation: Excess return was 1.4 points, information ratio 0.32, and capture 105% upside / 92% downside S2. Why it matters: Participation was favorable but only moderately efficient. Uncertainty: Only 72 monthly observations are available for both estimates. What would change the view: Downside capture above 100% or information ratio below zero.

06 Reliability and date-window sensitivity

All returned start dates preserved positive excess, but outcomes varied materially

ROLLING SHARPE RELIABILITY CHECK [S2]



The shorter window crossed below zero in 2022; the 36-month measure remained positive S2.

DATE-WINDOW SENSITIVITY DETAIL [S3]

START	END	MONTHS	MANAGER ANN. RETURN	BENCHMARK ANN. RETURN	EXCESS	SOURCE
2020-01-31	2025-12-31	72	6.5%	5.1%	+1.4pp	S3
2021-01-31	2025-12-31	60	6.2%	4.8%	+1.4pp	S3
2022-01-31	2025-12-31	48	4.8%	3.5%	+1.3pp	S3
2023-01-31	2025-12-31	36	12.0%	11.0%	+1.0pp	S3

Analysis — Stable sign, sensitive magnitude

Observation: Four windows had positive excess, while manager returns ranged from 4.8% to 12.0% S3. Why it matters: Relative sign was stable but absolute outcome varied. Uncertainty: The shortest window has 36 post-decline observations. What would change the view: A negative returned window or 36-month Sharpe below zero.

07 Benchmark sensitivity

The alternative benchmark preserves the sign but weakens the relative case

SECOND-BENCHMARK COMPARISON [S5]

BENCHMARK	STATUS	ANN. BENCHMARK RETURN	MANAGER EXCESS	INFORMATION RATIO	SOURCE
Northstar Core Equity Benchmark	Server matched	5.1%	+1.4pp	0.32	S1 / S2
Northstar Broad Equity Benchmark	Advisory	6.0%	+0.5pp	0.11	S5

Analysis — Benchmark choice changes strength, not sign

Observation: Excess return narrows from 1.4 percentage points to 0.5 percentage points and information ratio falls from 0.32 to 0.11 against the advisory alternative S2 S5. Why it matters: The relative case is less persuasive under a broader comparator. Uncertainty: The alternative benchmark is advisory rather than server matched. What would change the view: A canonical alternative benchmark with annualized excess at or below 0.0 percentage points.

08 Limitations and follow-ups

Four bounded gaps define the next diligence cycle

- The realized sample contains 72 monthly observations and one major drawdown episode, limiting inference about persistence S1.
- The primary benchmark is server matched, but mandate documentation was not included in the governed response S1.
- Capture ratios depend on returned up- and down-month classifications and may change as the sample extends S2.
- The 10-year expected-statistics outputs are model evidence and cannot be treated as realized manager performance S4.

01 **Can the manager provide mandate documentation confirming that the server-matched primary benchmark reflects the strategy actually managed during the full window?**
Benchmark fit determines whether the 1.4-point annualized excess return is decision-useful S1.

02 **What process or portfolio changes explain the -18.2% drawdown and subsequent recovery?**
The largest observed loss is the central absolute-risk event in the returned history S1.

03 **Can the manager reconcile gross, net, and composite return definitions for all 72 observations?**
Return-basis consistency is required before extending the performance conclusion to an investment vehicle S1.

04 **Would a governed refresh with at least 84 months preserve positive 36-month excess return and downside capture below 100%?**
Those thresholds would directly test persistence and defensive participation S2 S3.

A Appendix — Sources and methods

Saved sample responses and reproducibility lineage

Based on 5 of 5 evidence areas.

SERIES VIEW LINEAGE

SOURCE TAG	VIEW	PAYLOAD DIGEST
S2	cumulative_growth_base100	sha256:0101010101010101010101010101010101010101010101010101010101010101
S2	drawdown	sha256:0202020202020202020202020202020202020202020202020202020202020202
S2	rolling_return_12	sha256:0303030303030303030303030303030303030303030303030303030303030303
S2	rolling_return_36	sha256:0404040404040404040404040404040404040404040404040404040404040404
S2	rolling_excess_36	sha256:0505050505050505050505050505050505050505050505050505050505050505
S2	rolling_sharpe_12	sha256:0606060606060606060606060606060606060606060606060606060606060606
S2	rolling_sharpe_36	sha256:0707070707070707070707070707070707070707070707070707070707070707
S2	calendar_years	sha256:0808080808080808080808080808080808080808080808080808080808080808
S2	capture	sha256:0909090909090909090909090909090909090909090909090909090909090909
S2	distribution	sha256:0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a0a

Illustrative, Gradient Maintained — demo data, not the client's holdings or managers. All organizations, portfolios, managers and results in this public example are illustrative.

SOURCES

TAG	EVIDENCE	SOURCE	AS OF	CHECK
S1	Observed manager and primary-benchmark summary metrics	analyze_strategy_lab_compare	2025-12-31	Passed
S2	Realized paths, rolling statistics, relative metrics, capture, and distribution	run_strategy_lab_relative_return	2025-12-31	Passed
S3	Date-window robustness	run_strategy_lab_date_window_robustness	2025-12-31	Passed
S4	10-year simulated expected-statistics outputs	run_strategy_lab_expected_statistics	2025-12-31	Passed
S5	Alternative-benchmark sensitivity	analyze_strategy_lab_compare	2025-12-31	Passed

SOURCE AUDIT TRAIL

TAG	KEY PARAMETERS	DATA SCOPE	DIGEST
S1	manager=returns-test-northstar-select; benchmark=benchmark-test-northstar-core-equity; start=2020-01-31; end=2025-12-31	72 monthly observations; USD total return; illustrative sample data	sha256:aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa
S2	manager=returns-test-northstar-select; benchmark=benchmark-test-northstar-core-equity; rolling_months=12,36	Common 72-month manager/benchmark window; illustrative sample data	sha256:bbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb
S3	starts=2020-01-31,2021-01-31,2022-01-31, 2023-01-31; end=2025-12-31	Four returned start-date windows; illustrative sample data	sha256:ccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccc
S4	horizon=10y; outputs=annualized_return, annualized_volatility,terminal_wealth_median	Server-returned model outputs; illustrative sample data	sha256:ddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddd
S5	manager=returns-test-northstar-select; benchmark=benchmark-test-northstar-broad-equity; status=advisory	Same 72-month window; illustrative sample data	sha256:eeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee

Each digest is the SHA-256 fingerprint of the exact data the report used.